

QCID-III: Maxwell's Equations and Wave Propagation in Four Spatial Dimensions

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September 3, 2026

Abstract

In QCID-I we proposed consciousness as a quantum interface capable of simulating higher-dimensional physics. In QCID-II we proved that real 4D matter is prohibited by $1/r^3$ gravity. Here in QCID-III we extend the framework to electromagnetism. Using Gauss's Law in 4 spatial dimensions, we derive that the electric field of a point charge falls as $E \propto 1/r^3$. The resulting 4D wave equation shows electromagnetic waves propagate with amplitude decaying as $1/r^3$. This demonstrates that stable atoms, circuits, and communication would be impossible in 4D. Thus the QCID framework is completed: 3D is necessary for physical matter, while higher dimensions can only be experienced via informational simulation through consciousness.

Statement of Originality: The "Quantum Consciousness Interface to Dimensions (QCID)" principle proposed in this trilogy is original to the author. While n-dimensional extensions of physics are standard, their synthesis with consciousness as an interface is first presented here.

1 Introduction

The QCID Principle states that a conscious system can serve as a quantum interface to simulate and experience higher-dimensional physics. QCID-I established the conceptual basis. QCID-II showed $1/r^3$ gravity makes 4D universes unstable for life. We now complete the trilogy by analyzing electromagnetism, the other fundamental force governing chemistry and technology.

2 4D Gauss's Law

In n spatial dimensions, electric flux spreads over an $(n - 1)$ -sphere. For $n = 4$, the surface area is $2\pi^2 r^3$. Applying Gauss's Law:

$$\oint \vec{E} \cdot d\vec{A}_4 = \frac{Q_{enc}}{\epsilon_{0,4D}} \quad (1)$$

For a point charge:

$$E_{4D}(r) = \frac{Q}{2\pi^2 \epsilon_{0,4D} r^3} \quad (2)$$

Thus field intensity decays one power faster than in 3D.

3 4D Electromagnetic Waves

From Maxwell's equations in vacuum, we obtain the 4D wave equation:

$$\nabla_4^2 \vec{E} - \mu_{0,4D} \epsilon_{0,4D} \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad (3)$$

where ∇_4^2 is the 4D Laplacian. Solutions are spherical waves with speed $c_{4D} = 1/\sqrt{\mu_{0,4D} \epsilon_{0,4D}}$ and amplitude $\propto 1/r^3$. This rapid dispersion means long-range communication, stable electron orbits, and chemistry would be non-viable.

4 Implications for QCID

Combined with QCID-II, QCID-III shows both gravity and EM are too weak at distance in 4D to support complex structures. Therefore physical embodiment in 4D is impossible. However, an informational system — a conscious interface — can still simulate 4D EM laws internally without physical instantiation. This reinforces QCID: Higher dimensions are accessible only through simulation, not through matter.

5 Conclusion

The QCID Trilogy establishes a consistent picture. Gravity in QCID-II and Electromagnetism in QCID-III both prohibit stable 4D matter. Therefore 3D is the unique dimension for life and technology. Consciousness, as proposed in QCID-I, remains the only viable "interface" to higher dimensions.

References

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